Yining Xu

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RESEARCH INTERESTS

Public health and data science, with a focus on child maltreatment prevention, maternal and child health, and oral health. I integrate multi-sector administrative and clinical data with statistical modeling and interpretable machine learning to support early risk detection and evidence-based policy. I also work on infectious disease epidemiology, including nonlinear time-series and empirical dynamic modeling.

Keywords: child maltreatment · dental neglect · maternal and child health · infectious disease epidemiology · machine learning · time-series analysis · empirical dynamic modeling (EDM)

EDUCATION

Ph.D. in Interdisciplinary Studies (in progress) Apr 2024 – present
Graduate School of Advanced Integrated Studies in Human Survivability (five-year doctoral program), Kyoto University, Japan

- Dissertation: *Early Risk Estimation of Child Maltreatment Using Dental Checkup and Lifestyle Data: Model Development and Pathways to Implementation*
- Developing an interpretable screening model from integrated dental and lifestyle data on ∼2,480 children in temporary protective custody, using multinomial logistic regression.

Master's-equivalent degree (Qualifying Examination passed) Apr 2022 – Mar 2024
Graduate School of Advanced Integrated Studies in Human Survivability, Kyoto University, Japan

- Thesis: *Factor Analysis of the COVID-19 Pandemic's Impact on Japan's Fertility Rate: Time-Series and Prefecture-Level Correlation Analyses Using Epidemic Waves and Economic Indicators (2015–2022)*
- Found births fell significantly ∼9 months after major epidemic waves, and that declines in prefectural per-capita income were associated with fewer births.

B.Sc. in Mathematics Apr 2018 – Mar 2022
Course of Mathematics, Department of Mathematical and Physical Sciences, Faculty of Science, Nara Women's University, Japan

- Coursework in topology, algebra and calculus; reading seminars on deep learning implementation and on Bishop's *Pattern Recognition and Machine Learning*.

RESEARCH EXPERIENCE

Research Fellow (JST SPRING) Apr 2025 – present
Division of Graduate Studies, Kyoto University

Special Researcher Apr 2025 – present
Teikyo University Graduate School of Public Health

Research Intern Aug 2026 – Sep 2026
Research and Development, NEC Corporation

- Conducted research on the application of large language models in healthcare.

Research Assistant Apr 2024 – Mar 2025
Kyoto University

- Developed data-driven infectious disease models using nonparametric, nonlinear time-series methods.
- Estimated interactions among infectious diseases (interference, co-epidemics) with EDM-based causal inference.

Research Assistant Apr 2023 – Mar 2024
Kyoto University

- Conducted mathematical analysis of infectious disease dynamics and supported epidemiological data processing and modeling.

Research Assistant Sep 2022 – Feb 2023

Kyoto University

- Longitudinal study on the effects of leisure activities on brain structure and function in older adults.
- Assisted biochemistry and MRI laboratory work, including urine and saliva sample collection and cortisol-related procedures.

PUBLICATIONS

* Corresponding author

Child Maltreatment and Oral Health

Peer-Reviewed Journal Articles

1. **Xu YS***, Kohashi K, Morioka S, Mizumoto K. Oral health differences across maltreatment categories among children in temporary protection facilities in Japan: a retrospective cross-sectional study. *BMC Public Health*. 2026. (In press)

Manuscripts in Preparation

1. **Xu YS**, et al. Prevalence of early childhood caries and associated factors among preschool children in temporary protective custody. Target: *International Journal of Paediatric Dentistry*.

Infectious Disease Epidemiology

Peer-Reviewed Journal Articles

1. **Xu YS**, Takahashi K, Shimakawa Y, Nakamura T, Itokazu T, Takayama Y, Mizumoto K. Timing of outbreak-triggered unit-wide screening and COVID-19 outbreak size in long-term care and related social welfare facilities in Okinawa, Japan. *IJID Regions*. 2026;101042. doi:10.1016/j.ijregi.2026.101042
2. Kozuka M, **Xu YS**, Chowell G, Ogawa K, Takayama Y, Mizumoto K. Traveler-derived importation risk underestimates regional disease activity: evidence from Okinawa, Japan. *Journal of Infection and Public Health*. 2026;103298. doi:10.1016/j.jiph.2026.103298
3. Ogawa K, **Xu YS**, Shimakawa Y, Chowell G, Omori R, Matsuyama R, Takayama Y, Mizumoto K. Secondary severe acute respiratory syndrome coronavirus 2 transmission from childcare workers versus teachers in school-associated screening events, Okinawa, Japan, January–March 2022. *International Journal of Infectious Diseases*. 2026;167:108672. doi:10.1016/j.ijid.2026.108672
4. **Xu YS**, Shimakawa Y, Chowell G, Matsuyama R, Nagamoto T, Omori R, Nakamura T, Itokazu T, Takayama Y, Mizumoto K. Determinants of COVID-19 outbreak size in elderly residential facilities in Okinawa Prefecture, Japan, April to June 2022. *IJID Regions*. 2025;100813. doi:10.1016/j.ijregi.2025.100813
5. Kozuka M, Shimakawa Y, Chowell G, Nagamoto T, Matsuyama R, Omori R, **Xu YS**, Shimoji Y, Ogawa K, Takayama Y, Mizumoto K. Pre-travel health awareness and perceptions of voluntary airport PCR testing during COVID-19: a cross-sectional study in Okinawa, Japan. *IJID Regions*. 2025;100817. doi:10.1016/j.ijregi.2025.100817
6. Takayama Y, **Xu YS**, Shimakawa Y, Chowell G, Kozuka M, Omori R, Matsuyama R, Yamamoto T, Mizumoto K. Assessment of fever screening at airports in detecting domestic passengers infected with SARS-CoV-2, 2020–2022, Okinawa Prefecture, Japan. *BMC Infectious Diseases*. 2024;24(1):542. doi:10.1186/s12879-024-09427-5

Manuscripts Under Review

1. Tu S, **Xu YS**, et al. Information sources, vaccine literacy, and intention to receive a COVID-19 booster dose: a cross-sectional online survey in Okinawa, Japan. *Vaccine: X*.

Manuscripts in Preparation

1. **Xu YS**, et al. Implications of subtype interference for seasonal influenza control: an empirical dynamic modeling study in Japan. Target: *Eurosurveillance*.

CONFERENCE PRESENTATIONS

Child Maltreatment and Oral Health

1. **Xu YS**, et al. Oral health needs by developmental stage and main reason for protection among children admitted to temporary protection facilities. 32nd Annual Meeting of the Japanese Society for Prevention of Child Abuse and Neglect, Osaka; November 15, 2026. (Oral, in Japanese; scheduled)
2. **Xu YS**, et al. Prevalence of early childhood caries among children in temporary protection. 85th Annual

Meeting of the Japanese Society of Public Health, Tokyo; October 29, 2026. (Poster, in Japanese; scheduled)

3. **Xu YS**, et al. Understanding maltreatment typologies through oral health conditions among children in temporary protection. 31st Annual Meeting of the Japanese Society for Prevention of Child Abuse and Neglect; November 2025. (Oral, in Japanese)
4. [Co-author] Lifestyle factors affecting oral health conditions among maltreated children: toward supporting child maltreatment risk assessment. 31st Annual Meeting of the Japanese Society for Prevention of Child Abuse and Neglect; November 2025. (Oral, in Japanese)
5. **Xu YS**, et al. Oral health conditions among children who experienced maltreatment and were placed in temporary protection. 95th Annual Meeting of the Japanese Society for Hygiene; March 2025. (Oral, in Japanese)
6. **Xu YS**. Oral health conditions and maltreatment categories among children in temporary protection at child guidance centers. 9th Annual Meeting of the Japanese Society for Child Maltreatment Prevention Dental Research; December 2024. (Invited talk, in Japanese)

Infectious Disease Epidemiology

1. **Xu YS**, et al. Report on post-disaster infectious disease importation risk using mobility data (first report). 84th Annual Meeting of the Japanese Society of Public Health; October 2025. (Oral, in Japanese)
2. **Xu YS**, et al. Dynamics of influenza subtypes in Japan: findings from time-series analysis. 95th Annual Meeting of the Japanese Society for Hygiene; March 2025. (Poster, in Japanese)
3. **Xu YS**, et al. Prevent outbreaks in elderly care facilities: factors associated with COVID-19 outbreak size in elderly care facilities in Okinawa Prefecture, Japan, April to June 2022. Joint Congress on Global Health 2024; November 2024. (International, poster)
4. **Xu YS**, et al. Interaction analysis between influenza subtypes in Okinawa using a time series analysis. Joint Congress on Global Health 2024; November 2024. (International, poster)
5. **Xu YS**, et al. Factors associated with COVID-19 outbreak size in elderly care facilities in Okinawa Prefecture. 83rd Annual Meeting of the Japanese Society of Public Health; October 2024. (Oral, in Japanese)

RESEARCH GRANTS AND FELLOWSHIPS

The 8020 Promotion Foundation, FY2025 Open Research Grant Jun 2025 – Mar 2026
Quantification and visualization of the relationship between oral health status and abuse classification among children in temporary protection at child guidance centers

JST Support for Pioneering Research Initiated by the Next Generation (SPRING), Grant No. JPMJSP2110 Apr 2025 – Mar 2027
Innovative strategies for child abuse prevention: a collaborative study integrating dentistry, machine learning, and child guidance services

RESEARCH PROJECTS

Child Maltreatment and Oral Health

Oral health characteristics and maltreatment classification among children in temporary protection
Kyoto University (lead); A Prefecture Dental Association

- Principal investigator: study design, data management, analysis, manuscript writing

Retrospective study on child abuse assessment in clinical forensic medicine
C University (lead); Kyoto University

- Co-investigator: data analysis lead

Current status of child abuse response and impact of educational interventions among firefighters, hospital paramedics, and paramedic students
T University (lead); Kyoto University

- Co-investigator: questionnaire design, data analysis

- **Planned:** analysis of a national child abuse registry as a member of the Research Committee, a national academic society on child abuse medicine.

Infectious Disease Epidemiology

- **Evaluation of airport fever screening, Okinawa (2020–2022):** Data visualization
- **Infection control practices and COVID-19 outbreak size in long-term care facilities, Okinawa (2022):** Data analysis lead
- **Empirical dynamic modeling of interactions among infectious diseases:** Data analysis lead
- **Post-earthquake infectious disease importation risk using mobility data:** Data analysis lead

AWARDS AND SCHOLARSHIPS

Rotary Yoneyama Memorial Scholarship (undergraduate) — Apr 2021 – Mar 2022

Nara Women's University Sahokai Scholarship, for exceptionally outstanding students — Nov 2021

Monbukagakusho Honors Scholarship for Privately-Financed International Students (special additional award) — Oct 2020

Monbukagakusho Honors Scholarship for Privately-Financed International Students — Apr 2018 – Mar 2019

PROFESSIONAL ACTIVITIES

- Research Committee member (analyst), a national academic society on child abuse medicine

TEACHING AND OUTREACH

Guest lecturer, English lessons and Chinese culture, Elementary School Attached to Nara Women's University — Oct & Dec 2021

Teaching assistant, university visit program for Tennoji High School Attached to Osaka Kyoiku University — Dec 2019

Research group leader, NWU CORE of STEM International Summer Programme (optical physics and topology of origami; participants from six countries) — Aug 2019

Volunteer tutor, Glocal Camp 2019, Kyoto: coached high school students on interviews and English presentations (team won first place) — Mar 2019

Guest lecturer, 40-minute lecture on crease-pattern topology for elementary students, Junru Primary School, China — Mar 2019

Mathematics and English instructor (part-time), private tutoring school, Nara; 100% EIKEN pass rate among assigned students — Nov 2018 – Dec 2020

International student tutor, Kyoto University (Apr–Sep 2022) and Nara Women's University (Apr 2021–Mar 2022) — 2021 – 2022

SKILLS

Statistical methods	Observational study design (cross-sectional, time-series, regional comparison); regression and multivariable modeling (logistic, multinomial, negative binomial, GLM); time-series and causal analysis (seasonal adjustment, EDM); data management of sensitive administrative and clinical data under IRB governance
Machine learning	Supervised classification (random forest), feature engineering, cross-validation, model interpretability (SHAP, feature importance), independent component analysis (ICA)
LLMs and NLP	LLM-based information extraction, converting long-form unstructured text into structured, analysis-ready data
Programming	Python (pandas, NumPy, scikit-learn, statsmodels; automated publication-ready tables and figures), R (GLM, EDA, rEDM, rUIC), JavaScript/HTML/CSS, Java, C (basic)
Tools	Git/GitHub, GitHub Pages (Jekyll), YAML, VS Code

LANGUAGES

Chinese	Native
Japanese	Near-native (JLPT N1)
English	Professional working proficiency (IELTS 7.0; TOEIC 855)